

Jatin Madan

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University of California, Irvine — Master of Computer Science — GPA: 4.0 [Irvine, CA, USA] Sept 2023 - Dec 2024
Vellore Institute of Technology — B.Tech Computer Science and Engineering [Vellore, India] July 2016 - June 2020

SKILLS

Languages	Python, C#, PowerShell, Bash, SQL, Java, JavaScript, Swift
Security Research	Vulnerability Research, Threat Modeling, Exploit Development (PoC), Static & Dynamic Analysis (SAST/DAST), Application Penetration Testing, Code-Level Security Auditing, Design Review, Authentication & Security Protocols (OIDC, OAuth 2.0, SAML), Applied Cryptography, Supply Chain Security
Cloud & Infra	Azure (AKS, Key Vault, Entra ID, VNet, Function Apps, VMSS, Log Analytics), AWS (S3, Lambda), Multi-Cloud IAM, Kubernetes (AKS/GKE), Docker, Terraform, GitHub Actions, Azure DevOps, RBAC, Zero Trust, Network Segmentation
Tools	Microsoft Sentinel, Defender for Cloud, GHAS, Purview, Databricks, Power BI, Postman, PySpark, Pandas, NLTK, Django

EXPERIENCE

Deloitte — Engineering Manager — [Cloud Infra and Security Team](#) — [Seattle, WA](#) Jan 2025 - Present

- Conducted **security design reviews and threat modeling** across **multi-cloud, multi-tenant environments** — identifying **authentication, API, and identity attack vectors** across **2,000+ enterprise applications**, producing **proof-of-concept exploits** to demonstrate vulnerability impact and driving **systemic fixes** across affected application patterns
- Built and deployed an **MCP server on AKS** — performed **code-level security auditing** of the authentication layer, discovered **token handling and API authorization flaws**, and shipped **OIDC-based workload identity and Key Vault secrets management** fixes end-to-end, reducing identity attack surface by **60%**
- Developed **security tooling in Python and PowerShell** to automate **vulnerability detection, credential scanning, and RBAC misconfiguration analysis** across Entra ID tenants — prototyped, iterated, and deployed tools that reduced manual security review effort by **70%**
- Implemented **AI red-teaming and adversarial testing frameworks** — systematically probed **model guardrails, agent behaviors, and policy enforcement boundaries** via **Microsoft Purview and Defender for Cloud**, identifying exploitable gaps in AI system security posture across **5+ business units**
- Hardened **CI/CD pipelines** against supply chain attacks using **GHAS, SAST integration, signed attestations, dependency scanning, and automated policy enforcement** — assessed externally reported vulnerabilities for impact and shipped **systemic pipeline fixes** across **2,000+ application** deployments
- Partnered with **engineering, product, and infrastructure teams** to define **security expectations and secure-by-default development patterns**, conducting **static and dynamic code analysis** and translating findings into both individual fixes and **systemic remediation** for recurring vulnerability classes

ZS Associates — Associate Consultant Intern — [Cloud and Big Data Team](#) — [Philadelphia, PA](#) Jun 2024 - Aug 2024

- Performed **security auditing** of a large-scale Databricks environment — discovered **access control misconfigurations and privilege escalation paths** across 3,000+ datasets, developed **proof-of-concept access exploits** to demonstrate impact, and shipped end-to-end remediation via **Unity Catalog RBAC and encryption controls**
- Built a **security tool in Python** to **statically analyze Docker image configurations** for vulnerable packages and unsafe runtime patterns — automated triage across Databricks cluster images and reduced attack surface by eliminating runtime installation vectors (**14% runtime improvement**)
- Conducted **code-level review** of data pipeline infrastructure, identifying **insecure deserialization patterns and secret exposure risks**, and collaborated with the platform team to ship fixes and add **automated scanning** to the build pipeline

University of California, Irvine — Data & Tech Fellow — [Prof. Tom Andriola](#) — [Irvine, CA](#) Jan 2024 - Dec 2024

- Performed **threat modeling and vulnerability research** on a hybrid identity system (**Entra ID Connect, OIDC, Application Proxy**) — discovered **token replay, session hijacking, and lateral movement vectors**, developed **PoC attack scenarios** to demonstrate impact, and implemented cryptographic and policy-based fixes
- Built an **Apple Vision OS platform** in **Swift** with SSH tunneling and sandboxed execution — conducted **design review and dynamic analysis** of the credential management and network isolation layers, identifying and patching **authentication bypass and secret leakage risks** before deployment
- Prototyped a **static analysis tool** to scan authentication flows for **insecure token handling and misconfigured authorization policies**, triaging findings and producing **systemic fixes** adopted across the identity infrastructure

- Developed an **Automated Multi-Cloud Security Assessment Tool** in **Python** — performed **vulnerability discovery and analysis** across IAM, network, and RBAC layers of multi-tenant Azure environments, **prototyped exploits** for discovered misconfigurations, and drove **systemic fixes** reducing detection gaps by **35%**
- Conducted **application penetration testing and code-level auditing** of **PaaS databases and Kubernetes clusters** — discovered **privilege escalation, insecure deserialization, and lateral movement vulnerabilities**, and shipped **RBAC, secrets rotation, and identity lifecycle** fixes validated through re-testing
- Developed an **Azure DevOps IaC scanning extension in Python** performing **static analysis of Terraform templates** — detected **misconfigurations and supply chain policy violations** pre-deployment, integrated **automated triage and remediation workflows** end-to-end into CI/CD pipelines
- Performed **design review and threat modeling** of a **Secure SSO system and Video Conferencing Platform** — identified **OAuth 2.0 token handling flaws, API authorization gaps, and serialization vulnerabilities**, developed **PoC exploits**, and implemented **applied cryptography controls and network segmentation** fixes

CERTIFICATIONS		LINK TO ALL 11 CERTIFICATIONS	
● Microsoft Certified: Azure Security Engineer Associate 2021	AZ-500	H736-2347	Mar 28,
● Microsoft Certified: Azure Solutions Architect Expert 2021	AZ-304	H693-3978	Feb 21,
● Microsoft Certified: DevOps Engineer Expert 2021	AZ-400	H813-4148	May 16,
● Microsoft Certified: Azure Developer Associate 2021	AZ-204	H717-3092	Mar 13,
● Microsoft Certified: Azure Network Engineer Associate 2021	AZ-700	I483-3372	Nov 12,